MUKESH SETHY

MECHATRONICS ENGINEER

CONTACT

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EDUCATION

2013-2018
N.I.T ROURKELA

- Dual Degree M.Tech in Mechatronics and Automation and B.Tech in Mechanical Engineering

2010-2012
D.A.V PUBLIC SCHOOL, UNIT-8, BHUBANESWAR

- 12th, CBSE

TECHNICAL SKILLS

- Python
- MATLAB-Simulink
- Vector CANoe
- Vector CANape
- Solidworks/Fusion 360
- Eagle CAD/ KiCAD
- ROS 1

LANGUAGES

- English (Fluent)
- Hindi (Fluent)
- Odia (Fluent)
- German (Basics)

PROFILE

Robotics and Mechatronics Engineer with 8+ years of experience in robotics, embedded systems, and electric vehicle technologies. Skilled in C++/Python debugging, RTOS-based firmware, ROS/Linux development, and model-based design (MATLAB Simulink). Hands-on with Nvidia Orin Nano, ESP32, STM32, and PCB/electronics design. Proven leadership in L3 support, ECU lifecycle management, and PLM/BoM process optimization. Strong academic foundation with dual degrees in Mechatronics and Mechanical Engineering, complemented by publications, a patent in EV energy management, and leadership in robotics competitions.

WORK EXPERIENCE

Rapyuta Robotics, Chennai <->Tokyo

Electrical Engineer — JUNE 2025 - PRESENT

- Leading L3 support for mechatronics field issues in ASRS robots, ensuring rapid resolution and reliability.
- Debugging embedded systems and firmware in C++/Python across mobile robots, battery exchangers, elevators, and other subsystems.
- Developing solutions on Nvidia Orin Nano (Linux, ROS) and implementing RTOS-based firmware on ESP32/STM32 microcontrollers for real-time robotics applications, while driving new ASRS functions and conducting multi-site root cause analysis (RCA).
- Contributing to PCB development and electronics design, supporting hardware integration and system reliability.
- Heading PLM and BoM part numbering systems, streamlining collaboration between SCM and engineering teams.
- Driving core mechatronics development for ASRS robots, focusing on system enhancements and reliability improvements.

Daimler Trucks Innovation Centre India - Split from Mercedes Benz Research and Development India - Bengaluru

AUGUST 2018 - JUNE 2025

Senior Engineer

- Component Responsible (ECU Owner) for Cooling Control Unit ECU.
- Managed new ECU lifecycle from specification development to Start of Production.
- Designed ECU I/O wiring harness for CAN, sensors and actuator communications.
- Supported Cybersecurity (TARA) and Functional Safety (HARA) assessments.
- Performed root cause analysis for production and component issues.
- Implemented country and customer specific cooling system requirements.
- Built a thermal test bench with actual bus components for validation.

Engineer - Data Analytics for e-Citaro(electric Bus)

- Imported and processed logged system data from e-Citaro dataloggers in field vehicles for system functionality monitoring.
- Developed a Python-based GUI for CAN signal data visualization.
- Monitored the functionality of cooling and HVAC systems for high-voltage components.
- Supported MIL/SIL/HIL testing for Cooling System application software development.

ACADEMIC PROJECTS

Path Planning for NAO Humanoid Robot 2017-2018

- Master Thesis: Developed AI-based robot localization, obstacle mapping, and path planning using RGBD camera data.

Automatic Launch Control System 2016

- Designed a control system to reduce wheel slippage during vehicle launch using CAN Bus commands.

Pneumatic Gear Shifting System 2015-2016

- Designed a semi-automatic gear-shifting system controlled by microcontroller and ECU.

Formula SAE: Data Acquisition System 2015-2016

- Designed a semi-automatic gear-shifting system controlled by microcontroller and ECU.

Balloon Satellite Telemetry and Tracking 2014

- Led electronics team to build a satellite carrying sensors and cameras for atmospheric data logging and recovery.

PUBLICATIONS AND PATENT

Publications:

- "A Sensor-based Computer Vision Approach for Humanoid Navigation" – ICN:3I-2017, IIT Roorkee.
- "Humanoid Navigation: An Intelligent Computer Vision-Based Approach" – iEECON2018.
- "An Intelligent Computer Vision Integrated Regression-Based Navigation Approach for Humanoids."

Patent:

- System for Optimal Management of Energy Consumption in Electric Vehicles (Application Number: 202211025514).

EXTRA CURRICULAR ACTIVITIES

Captain 2014-2015
CYBORG - Robotics and Automation Club of NIT Rourkela

Electronics Head 2014-2015
ECO-KART - Fabrication of a motor powered Go-Kart

Vice-Captain and Electronics Head 2015-2017
FORMULA SAE - Fabrication of a Formula Student vehicle

First Position 2014
IRC 2014, International Robotics Challenge (Zonal Round)